

Early Fault Detection & Risk-Based Maintenance Prioritization

Multi-Farm SCADA Data Analytics & Deep Learning for Wind Turbines

Team Name: Team Sigmoid | Project Code: DLE-AI-202

Authors: Ziyad Muradov, Ismayil Yusifli, Emin Huseynli, Kamal Soltanaliyev

Verification Method: Git History Audit (shortlog & log --author)

1. INDIVIDUAL CONTRIBUTION MATRIX

MEMBER	CODE & INFRASTRUCTURE	REPORT SECTIONS	EXPERIMENTS & ANALYSES	OTHER RESPONSIBILITIES
Ziyad Muradov	GRU model design, training loop, evaluation script (src/model/); run_all.sh ; requirements.txt and repo scaffold; diagnosed and fixed CSV-delimiter bug, grid_power_factor threshold unit-mismatch bug, and scaler-precision issue in Farm C data pipeline.	Method	Trained and tuned the GRU across 6 configurations (learning rate, dropout, feature count, architecture) on Farm A and Farm C; ran 3-way baseline comparison.	Project architecture scaffold & repository structure setup.
Ismayil Yusifli	Data acquisition, cleaning, and quality audit for Farm A and Farm C (src/data/); physical feature engineering and cross-farm sensor mapping (configs/physical_sensor_mapping.yaml); baseline models (src/baseline/) for both farms; Farm A/B re-export using frozen Farm-C scaler.	Experimental Setup, Related Work	Built and ran Logistic Regression and HistGradientBoosting baselines on Farm A and Farm C.	Documented physical-feature domain-generalization contract (FEATURE_PIVOT_NOTE.md , SERVER_RUNBOOK.md).
Emin Huseynli	Evaluation suite (src/eval/metrics.py , plots.py): lead-time metric, per-turbine/per-farm breakdown, cost-sensitive metric, calibration analysis; deployment note and Dockerfile; found and fixed hardcoded input-size bug in deployment benchmark.	Results	Cross-farm Task 15 (results table) and Task 16 (distribution-shift analysis, compression-factor method); independently re-verified Tasks 17/18 with own implementation.	Identified Farm-A/11-feature checkpoint mismatch before report inclusion; completed cross-farm Tasks 17 (label shift) & 18 (turbine-model shift) analysis, originally delegated to Emin.
Kamal Soltanaliyev	Prioritization layer (src/prioritization/): core formula, degradation-rate calculation, prediction aggregation, criticality scoring, AccessCost model; full pipeline integration and tests; cross-farm generalization experiment execution and CLI.	Discussion & Limitations	Ran Farm-C-to-Farm-A/B cross-farm generalization experiment.	Delegated Tasks 17–18 to Emin; co-developed CLI execution interface.

2. DISCLOSED GIT-IDENTITY ISSUES

The team worked from a shared JupyterLab machine for parts of the project. Multiple times, a person's local git identity was not reset before another person committed, causing commits to be attributed to the wrong team member. Each case below was found via git shortlog -sn --all and git log --author=..., cross-checked against commit content and branch names.

- Ziyad's commits split across two identities:** Ziyad Muradov (10 commits) and TheDarkString (15 commits) — caused by a local git config change partway through the project. Combined: 25 commits, all verified by content to be Ziyad's work.
- Emin's commits split across two identities:** Emin Huseynli and ehuseynli96-ops — same root cause. Combined total correctly reflects Emin's evaluation-suite work.
- Commit misattribution (Ziyad under Emin):** One commit authored by Ziyad appeared under Emin's identity (the Farm-C grid_power_factor threshold fix) due to the shared machine having Emin's git config active. Disclosed here and left unrewritten, as subsequent team work had already been built on top of it.
- Commit misattribution & rebase correction:** 4 commits authored by Emin appeared under Ismayil's identity (Tasks 15–16 results table & distribution-shift analysis). Found, verified by content (branch names, task numbering), and corrected via interactive rebase before further work was built on top. Now correctly attributed to Emin Huseynli.
- Historical commit patterns:** A small number of Ismayil's earlier commits show wording patterns similar to Emin's known task areas. Investigated but left uncorrected to avoid rewriting a large portion of subsequent shared repository history. Flagged here for full transparency.

3. CONFIRMATION & SIGN-OFF

By signing below, each member confirms this report accurately reflects their contribution to the project.

Ziyad Muradov

Signature: *Ziyad Muradov*
Date: 9 September, 2026

Ismayil Yusifli

Signature: *Ismayil Yusifli*
Date: 9 September, 2026

Emin Huseynli

Signature: *Emin Huseynli*
Date: 9 September, 2026

Kamal Soltanaliyev

Signature: *Kamal Soltanaliyev*
Date: 9 September, 2026